1) $A = 360{,}56251_d$ $\qquad B = -10{,}46875_d$

a) $360{,}56251_d$

$360_d = 0001\ 0110\ 1000_b = 0x168 = 1{,}0110\ 1000 \cdot 2^8$

| | | | |
|---|---|---|---|
| $0{,}56251 \cdot 2 = 1{,}12502$ | 1 | 1 | $\rightarrow 2^{-1}$ |
| $0{,}12502 \cdot 2 = 0{,}25004$ | 0 | 2 | |
| $0{,}25004 \cdot 2 = 0{,}50008$ | 0 | 3 | |
| $0{,}50008 \cdot 2 = 1{,}00016$ | 1 | 4 | $\rightarrow 2^{-4}$ |
| $0{,}00016 \cdot 2 = 0{,}00032$ | 0 | 5 | |
| $0{,}00032 \cdot 2 = 0{,}00064$ | 0 | 6 | |
| $0{,}00064 \cdot 2 = 0{,}00128$ | 0 | 7 | |
| $0{,}00128 \cdot 2 = 0{,}00256$ | 0 | 8 | |
| $0{,}00256 \cdot 2 = 0{,}00512$ | 0 | 9 | |
| $0{,}00512 \cdot 2 = 0{,}01024$ | 0 | 10 | |
| $0{,}01024 \cdot 2 = 0{,}02048$ | 0 | 11 | |
| $0{,}02048 \cdot 2 = 0{,}04096$ | 0 | 12 | |
| $0{,}04096 \cdot 2 = 0{,}08192$ | 0 | 13 | |
| $0{,}08192 \cdot 2 = 0{,}16384$ | 0 | 14 | |
| $0{,}16384 \cdot 2 = 0{,}32768$ | 0 | 15 | |
| $0{,}32768 \cdot 2 = 0{,}65536$ | 0 | 16 | |

$A = 1{,}0110\ 1000\ 1001\ 0000\ 0000\ 000_b \cdot 2^8$

$S = 0 \qquad e = 8 \qquad E = 127 + e = 135_d = 1000\ 0111_b$

$M = 0110\ 1000\ 1001\ 0000\ 0000\ 000$

$A = \underbrace{0}\ \underbrace{1000}_{4}\ \underbrace{0111}_{3}\ \underbrace{0110}_{B}\ \underbrace{1000}_{4}\ \underbrace{1001}_{4}\ \underbrace{0000}_{8}\ \underbrace{0000}_{0}\ \underbrace{000}_{0} = 0x\ 43B44800$

Fehler: $\left|360{,}56251 - \left(360 + 2^{-1} + 2^{-4}\right)\right| = \left|360{,}56251 - 360{,}5625\right| = 10^{-5}$

$B = -10{,}46875 \quad \rightarrow S = 1 \qquad 10_d = 1010_b = 1{,}010 \cdot 2^3$

| | | | |
|---|---|---|---|
| $0{,}46875 \cdot 2 =$ | $0{,}9375$ | 0 | 1 |
| $0{,}9375 \cdot 2 =$ | $1{,}875$ | 1 | 2 |
| $0{,}875 \cdot 2 =$ | $1{,}75$ | 1 | 3 |
| $0{,}75 \cdot 2 =$ | $1{,}5$ | 1 | 4 |
| $0{,}5 \cdot 2 =$ | $1$ | 1 | 5 |

$M = 0100\ 1111\ 0000\ 0000\ 0000\ 000$

$B = -1{,}01001111 \cdot 2^3$

$e = 3 \rightarrow E = 127 + e = 130 \qquad S = 1$

$$B = \underbrace{1}_{C}\ \underbrace{1000}_{1}\ \underbrace{0010}_{2}\ \underbrace{0100}_{7}\ \underbrace{1111}_{8}\ \underbrace{0000}_{0}\ \underbrace{0000}_{0}\ \underbrace{0000}_{0}\ \underbrace{000}_{0} = 0x\,C1278000$$

(S: 1; E: 1000 0010; M: 0100 1111 0000 0000 0000 000)

Der Fehler ist 0, da sich die Mantissen-Berechnung ohne Rest ausgegangen ist.

b) $C = 0x\,BF7C0000 =$

$= \underbrace{1}_{S}\underbrace{011\ 1111\ 0}_{E}\underbrace{111\ 1100\ 0000\ 0000\ 0000\ 0000}_{M}{}_b$

$S = 1 \qquad E = 0111\ 1110_b = 126_d \quad e = 126 - 127 = -1$

$C = -1{,}11111_b \cdot 2^{-1} = -0{,}1111\ 1100_b = -\left(2^{-1} + 2^{-2} + 2^{-3} + 2^{-4} + 2^{-5} + 2^{-6}\right) = -\frac{63}{64_d}$

$C = -0{,}984\,375_d$

$D = 0x\,44302000$

$$D = \underbrace{0}_{S}\,\underbrace{100\ 0100}_{E}\ \underbrace{0011\ 0000\ 0010\ 0000\ 0000}_{M}$$

$S = 0$

$E = 1000\ 1000_b = 136_d \qquad e = E - 127 = 136 - 127 = 9$

$D = 1{,}0110\ 0000\ 0100_b \cdot 2^9 = 0010\ 1100\ 0000{,}1_b = 2^9 + 2^7 + 2^6 + 2^{-1} = 704{,}5_d$

c) $C \cdot D$

·) Sign-Bit berechnen: $S_C \neq S_D \Rightarrow S = 1$

·) Exponent berechnen:

$$M_C \cdot 2^{E_C} \cdot M_D \cdot 2^{E_D}$$

$E = E_C + E_D = 126 + 136 - 127 = 135$

·) Mantisse berechnen:

```
  1,111111 . 1,0110000001
  ----------------------
  111111000000000000
     11111100000000000
      1111110000000000
       111111000000000
                111111
  ----------------------
  10,1011010101111111
```

·) Exponent korrigieren:

$E = 135 + 1 = 136_d = 1000\ 1000_b$

·) Befüllen: $M = 0101\ 1010\ 1011\ 1111\ 0000\ 000$

$$C \cdot D = \overset{S}{\underbrace{1}}\overset{E}{\underbrace{100\ 0100}}\ \overset{M}{\underbrace{0010\ 1101\ 0101\ 1111\ 1000\ 0000}} = 0x\,C42D5F80$$

C 4 2 D 5 F 8 0

·) Probe: $-1{,}0101\ 1010\ 1011\ 1111\ 0000\ 000 \cdot 2^{136-127} =$

$= -0010\ 1011\ 0101{,}0111\ 1110_b = -(2^0 + 2^2 + 2^4 + 2^5 + 2^7 + 2^9 + 2^{-2} + 2^{-3} + 2^{-4} + 2^{-5} + 2^{-6} + 2^{-7}) =$

$= -693{,}492187\,5_d \overset{\checkmark}{=} 704{,}5 \cdot (-0{,}984375)$

d) C + D :

•) Exponent anpassen:

$$-1{,}11111 \cdot 2^{-1} + 1{,}0110\ 0000\ 0100 \cdot 2^{9} =$$

$$= -0{,}0000\ 0000\ 0111\ 1110 \cdot 2^{9} + 1{,}0110\ 0000\ 0100 \cdot 2^{9}$$

•) Mantissen addieren:

$$\begin{array}{r} 1{,}0110\ 0000\ 0100\ 000 \\ -\ \underline{0{,}0000\ 0000\ 0111\ 111} \\ 1{,}0101\ 1111\ 1100\ 001 \end{array}$$

•) Befüllen: $S = 0$ $\quad E = 127 + 9 = 136_d = 1000\ 1000_b$

$M = 0101\ 1111\ 1100\ 0010\ 0000\ 000$

$$C + D = \underbrace{0100}_{4}\ \underbrace{0100}_{4}\ \underbrace{0010}_{2}\ \underbrace{1111}_{F}\ \underbrace{1110}_{E}\ \underbrace{0001}_{1}\ \underbrace{0000}_{0}\ \underbrace{0000}_{0}{}_b = 0x\ 442FE100$$

•) Probe:

$$C + D = -0{,}984375 + 704{,}5 = 703{,}515625$$

$$1{,}0101\ 1111\ 1100\ 0010\ 0000\ 0000_b \cdot 2^{9} =$$

$$= 0010\ 1011\ 1111{,}1000\ 0100_b =$$

$$= 2^0 + 2^1 + 2^2 + 2^3 + 2^4 + 2^5 + 2^7 + 2^9 + 2^{-1} + 2^{-6} = 703{,}515625$$

$=$ ✓